

ERP Match Pro

3-Way Matching Portal — Project Explainer

A plain-English guide to what this project is, the real-world problem it solves, how it works step by step, and how to run and demo it.

Prepared for: Vishvajit | Internship project

1. What Problem Does This Solve?

When a company buys something from a vendor, three separate documents get created along the way:

- **Purchase Order (PO)** — the document the company creates saying "we want to buy 100 units of X at ■1,000 each from this vendor." This is the company's own record of what it agreed to buy.
- **GRN (Goods Receipt Note)** — created when the goods physically arrive at the warehouse, confirming what was actually received. This matters because sometimes a vendor ships less than what was ordered, ships the wrong thing, or ships late.
- **Invoice** — the bill the vendor sends afterward, asking to be paid.

Before the company's **Accounts Payable (AP)** team — the department that pays vendors — approves payment on that bill, they are supposed to check three things: did we actually order this (PO)? did we actually receive it (GRN)? and does the vendor's bill charge the price we agreed on? If all three agree, it is safe to pay. If any of them disagree — wrong price, wrong quantity, something never received — that is a red flag someone needs to catch before money leaves the company.

Doing this by hand for every single invoice — opening three separate documents and comparing numbers — is slow and error-prone, especially for a company processing hundreds of invoices a month. This project automates that check.

2. What Is “3-Way Matching”?

It's the name for the process of comparing the PO, the GRN, and the Invoice against each other — specifically the price — before approving payment. This is a standard fraud- and error-prevention step used by real finance departments everywhere, not something specific to this project.

Worked example (from the original assignment spec):

Field	PO	GRN	Invoice	Result
Price	1000	1000	1000	MATCH → auto-export
Price	1000	1000	1050	MISMATCH → ask user to correct

3. How the App Works, Step by Step

- **1. Login** — the AP employee signs into the portal.
- **2. Upload** — they upload a vendor invoice (a PDF or a photo/scan of one).
- **3. OCR extraction** — OCR (Optical Character Recognition — the technology that reads text out of an image, the same idea as how a phone camera can scan a business card) automatically pulls out the invoice number, vendor name, PO number, GRN number, price, and other fields, so nobody has to type them in by hand.
- **4. Fetch PO & GRN** — the app looks up the matching PO and GRN inside Odoo. Odoo is an **ERP** (Enterprise Resource Planning) system — the software a company uses as its central system of record for purchasing, inventory, and accounting. In a real company, the PO and GRN already live in Odoo because that's where purchasing happens.

- **5. Compare** — the app checks whether the price agrees across all three documents.
- **6a. If it matches** — the invoice is automatically approved, exported into Odoo as a “Vendor Bill” (ready to be paid), and the whole event is saved to a permanent history log. No human needed.
- **6b. If it doesn't match** — the app highlights exactly what's different and asks the employee to either correct the value and give a reason (which is permanently logged — auditors want to know why a number was changed), or leave it marked “Pending Review” so it definitely does not get paid until someone resolves it. A correction automatically re-runs the check.
- **7. Reporting** — anyone can filter processed invoices by date, vendor, PO number, or price, and download the results as an Excel/CSV/PDF report for record-keeping.

4. What Each Screen in the App Does

Screen	What it's for
Dashboard	Overview: how many invoices processed, matched, pending, mismatched today; trend chart; recent activity
Upload Invoice	Where you upload a file (or paste text) and OCR pulls out the fields for you to review.
Invoice Queue	A flat worklist of everything currently in the pipeline, with quick status and an "Open" button.
3-Way Matching	The comparison screen: shows PO vs GRN vs Invoice side by side with green/red indicators, and the ability to mark mismatches.
Review Center	Search, filter, and bulk-act on invoices; view an individual invoice's full audit trail (who did what, when)
Reports	Filter processed invoices and export Excel / CSV / PDF.
Analytics	Deeper charts: which vendors have the best match rate, mismatch reasons, how long invoices take to process
Settings	Your profile, the Odoo connection (and a "Test Connection" button), dark mode, and (for Admins) user permissions

5. Technology Used, and Why

- **Python + Streamlit** — Streamlit is a tool for building web apps quickly using only Python, no separate frontend coding language needed. This matches what the assignment specifically asked for.
- **SQLite database** — a simple, file-based database that needs no separate server or installation step, used to store invoices, users, and the audit history.
- **Odoo (mock + live modes)** — since you don't yet have real Odoo login credentials, the app currently runs in "mock" mode: it simulates Odoo using realistic sample purchase orders and receipts, so the whole workflow works end-to-end right now. It's built so that once you get real credentials, you change a setting (not the code) to point it at the real Odoo server.
- **EasyOCR** — the OCR engine used to read invoice fields from uploaded files. There's also a manual "paste text" option so you can test the field-extraction logic without installing the (large) OCR library.

6. Try It Yourself

On the Upload Invoice page, use the "Paste invoice text" tab (no OCR install needed) and try these two examples:

Example A — a clean match (auto-exports):

```
Invoice No: INV-2024-101
Vendor: Tech Supplies Inc
PO No: PO-2024-001
GRN No: GRN-2024-001
Qty: 100
Unit Price: 1000
GST Amount: 18000
Grand Total: 118000
Currency: INR
```

Example B — a price mismatch (needs correction):

```
Invoice No: INV-2024-102
Vendor: Office Products Ltd
PO No: PO-2024-002
GRN No: GRN-2024-002
Qty: 50
Unit Price: 1600
GST Amount: 13500
Grand Total: 93500
Currency: INR
```

PO-2024-002's real price is ₹88,500, so Example B will show a mismatch. Enter 88500 as the corrected price with a reason, and it will re-check and auto-export.

7. How to Run It

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
streamlit run app.py
```

Then open the local URL it prints (usually <http://localhost:8501>) and log in with:

```
admin@erpmatchpro.com / admin123 (Admin)
ap@erpmatchpro.com / ap12345 (Accounts Payable)
manager@erpmatchpro.com / mgr12345 (Manager)
```

8. Glossary

Term	Meaning
AP (Accounts Payable)	The department responsible for paying the company's vendors/suppliers.
PO (Purchase Order)	The company's own record of what it agreed to buy, at what price.
GRN (Goods Receipt Note)	The record created when ordered goods physically arrive.
Invoice	The bill the vendor sends asking to be paid.
3-Way Matching	Comparing PO, GRN, and Invoice before approving payment.
ERP	Enterprise Resource Planning — a company's central software system for purchasing, inventory,
Odoo	A specific, widely-used ERP software product.
OCR	Optical Character Recognition — technology that reads text out of images/scans.
Streamlit	A Python tool for building simple web apps without writing separate frontend code.
Audit Trail	A permanent, timestamped log of every action taken on a record, kept for compliance.
Mock mode	A simulated version of an external system (here, Odoo) used for testing/demoing without a real c